

Hydrodealkylation (HDA) of Toluene

CENG 113, SP 2026

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This project is a continuation of the CENG 100 and 102 project concerning the hydrodealkylation (HDA) of toluene to form benzene.¹

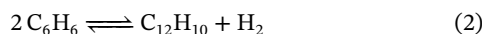
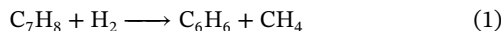
Process description

The production process is shown in fig. 1. Fresh toluene and hydrogen at 30 and 50 mol·s⁻¹ combine with toluene and light gas (H₂, CH₄) recycle streams. The reactor feed (stream 3, 250 °F) is preheated with reactor effluent (H-101), then heated to 1150 F with a fired heater (H-102), before entering a tube reactor (R-101). Hot reactor effluent is used to preheat the reactor feed and then proceeds to a series of separators. The process heat exchanger (H-101) is to be modeled as a counterflow PFR without reaction; the fired preheater (H-102) burns CH₄ to ensure the reactor feed is always 1150 °F.

Downstream of the reactor, a flash drum (V-101) recovers all H₂ and 95% of all CH₄ fed in stream 7. Most of the light gas effluent is recycled to the process feed in stream 9 with some purged in stream 10. A series of distillation columns (T-101 through T-103) separate all remaining components, with toluene recycled from T-103 to join the process feed in stream 16. All distillation columns are assumed to operate with perfect splits of the light key (top component).

Kinetics

The gas-phase thermal reactions are



with rates

$$r_1 = 3.686 \times 10^6 \exp\left(\frac{-25616}{T}\right) p_T p_H^{1/2} \quad (3)$$

$$r_2 = 5.987 \times 10^4 \exp\left(\frac{-25616}{T}\right) [p_B^2 - 4.264 p_D p_H] \quad (4)$$

where r_1 and r_2 have units of lbmol·min⁻¹·ft⁻³, T is the absolute temperature in Kelvin, and all partial pressures p_i are in psia.

¹You do not need to have completed that project to complete this one, nor have access to it.

Problem 1

Suppose reaction 1 was reversible and neglect the side reaction. Estimate equilibrium conversion at 1150 °F if the feed contains only equal parts hydrogen and methane, and 10 mol% toluene. Briefly evaluate the validity of assuming reaction 1 to be irreversible. Should complete conversion also be assumed based on these results?

Problem 2

Create a pseudo-Levenspiel plot based on reaction 1 only, assuming the toluene reactor feed is 1 (any units). Which reactor, PFR or CSTR, should be used and why?

Problem 3

Assume R-101 is a PFR operated isothermally and isobarically at 1150 °F and 500 psia, and a single pass toluene conversion of 75%. Vary the light gas recycle ratio $R = n_9/n_{10}$ from 0 to 20 to create a 1-by-3 plot of reactor volume, benzene production rate, and diphenyl production rate. Use these plots to justify a recommendation of $R = 10$.

Problem 4

Assume $R = 10$ and the reactor is adiabatic with pressure drop. Create a 1-by-3 figure containing profiles of all species molar flow rates (left), temperature (middle), and pressure (right) as a function of **reactor length, L , in feet**. Reactor diameter is 10 ft; assume $\mu = 4 \times 10^{-5}$ Pa·s (constant).

Problem 5

For the previous question, model heat exchanger H-101 as 150 identical, counterflow “reactors” but with no reaction occurring and no pressure drop. Each tube has a diameter of 3 cm and all tubes together have a total surface area of 700 m². Stream 3 is uniformly divided among and flows within the 150 tubes; similarly Stream 6 flows outside the tube. Create a plot of T_3 and T_6 as a function of **tube length, L , in feet**.

Estimate the CO₂ production rate (in metric tons per year) from H-102 if heating is provided by complete combustion of CH₄ with a stoichiometric quantity of O₂. Estimate approximately how many vehicles this is equivalent to. (*hint: compare methane’s enthalpy of combustion to the energy needed to heat stream 4 from T_4 to T_5*)

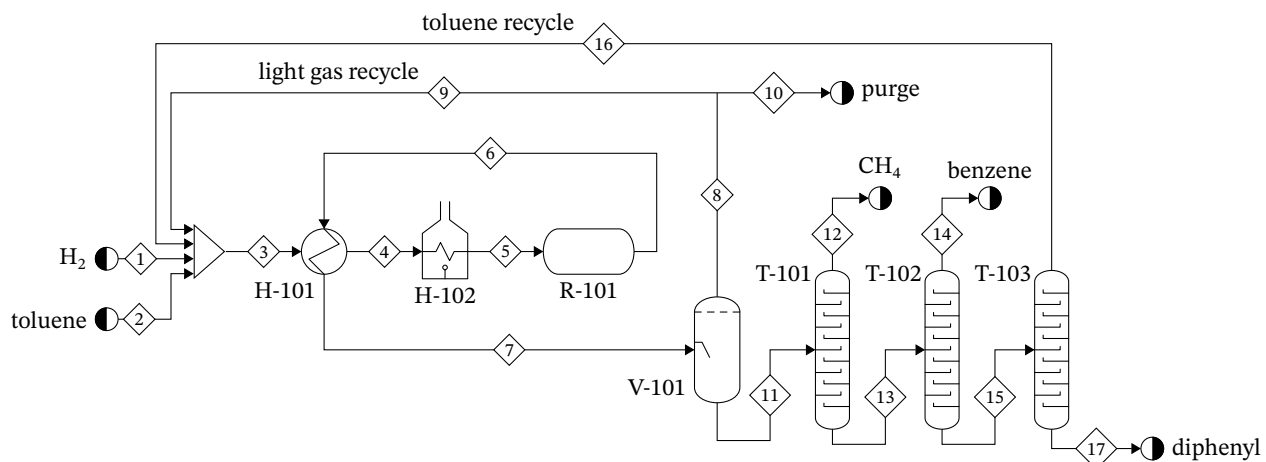


Figure 1: Simplified toluene HDA process, after Luyben (1998).